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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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1. Appointing an Independent Ethics Advisor

As part of the requirement of the SymAware project, we reached out to several potential ethics advisors (EA). We contacted a number of professionals in this field and received responses from a subset of them. After careful consideration and evaluation, we ultimately made the decision to select Mr. Matthijs Smakman as our EA for the project.

Matthijs is currently employed as an Associate Professor at the Institute for ICT, HU University of Applied Sciences Utrecht and at the Department of Communication Sciences, Faculty of Social Sciences, Vrije Universiteit Amsterdam. His expertise lies in the field of ethics, and social robotics. He was involved in the European L2TOR project as a visiting scholar regarding the ethical implications of social robots for second language learning during the final symposium in 2018 in Istanbul. His PhD thesis is specifically focused on implementing robot tutors in a morally justified way. His Google scholar contains the relevant publications:

Matthijs Smakman's Publications

His specialty is an ethics by design, values sensitive design approach. Furthermore, on a national level, he is a scientific advisor for a robotic company developing storytelling robots for children. Lately, he is involved as an ethics researcher in two national projects focused on AI in healthcare.

As part of our collaboration with Matthijs we will be working closely with him on the ethics deliverables of the project. We value his expertise and will seek his advice on any project activity that may have ethical implications. We are thrilled to have him on board as the EA for our project and we are confident in his ability to provide valuable guidance and insight on ethical matters related to the project. We are confident that with Matthijs's guidance, we will be able to meet the ethical deliverables and uphold the highest ethical standards throughout the project. The curriculum vitae is attached below this document.

Curriculum Vitae Matthijs Smakman

Personal information

Name:	Matthijs H. J. Smakman
Address:	Bankstraat 58 3581 LP Utrecht The Netherlands (NL)
Tel:	+31 (0)6 283 098 74
E-mail:	mhjsmakman@gmail.com
Place of birth - date:	Alphen aan den Rijn (NL), 06-09-1988
Nationality:	Dutch

Employment history

2022 – present

Associate professor (Hogeschoolhoofddocent) at Institute for ICT, HU University of Applied Sciences Utrecht
Responsible for combining research and education, end responsible for ICT curriculum, examination, internalization and the student excellence program.

Research Chair Digital Innovations in Healthcare at Research Group Digital Ethics
Responsible for the acquisition of research grants, project management, valorization, network building, and research.

2021 – present

Lecturer at Department of Communication Sciences, Faculty of Social Sciences, Vrije Universiteit Amsterdam
Responsible for master thesis supervision and lecturing on Human-Robot Interaction.

2014 - 2021

Lecturer at Institute for ICT, HU University of Applied Sciences Utrecht
Teaching bachelor and master students and developing courses in the fields of: Social Robotics, Business Process Management, Ethics by Design, Privacy, Law and Security, ICT and organizations, , Process Analysis, and thesis supervision.

Researcher at Research Group Digital Ethics, HU University of Applied Sciences Utrecht
Researching the social and ethical implications of new technologies, such as: robotics, quantified self, personalized content, AI and wearables.

2012 - 2014

IT-consultant at You-Get Business Services Provider
Analyzing, optimizing and implementing business processes management systems (BPMS) at several commercial companies in the educational sector.

2013 - 2014, Business Development manager
Setting up a corporate start-up (MVO Masters) that supports organizations with questions regarding Corporate Social Responsibility (CSR). Developing IT tools to integrate CSR in business processes in a measurable way.

Ancillary functions and memberships

2023 – present	**Special member, Innovatienetwerk Ouderenzorg Utrecht**
2021 – present	**Chair, International Social Robotics Contest**
2020 – present	**Scientific advisor, SAMbuddy Vertelknuffel**
2019 – present	**Member, KNVI**

Education

2017 - 2023
PhD

Department of Communication Sciences[1] Faculty of Social Sciences, Vrije Universiteit Amsterdam
Developing an empirically validated theory and guideline for applying robot tutors in education, supervised by prof. dr. Elly Konijn.
Dissertation approved, defense planned on 31 May 2023.

2015 - 2016
Master

Philosophy, Business and Organization Ethics, Tilburg University
Specialized in philosophical questions concerning technical innovations, such as: robotics, quantified self, personalized content and wearable technology.

2008 - 2012
HBO, Bachelor

Industrial Engineering & Management, HU University of Applied Sciences Utrecht

Certificates

2017 - 2018	**Basis Kwalificatie Onderwijs Ontwerpen**
2016 - 2017	**Basis Kwalificatie Examinering (BKE)**

Grants and awards received

2022 – 2023
Robotics for Sutainable Development Goals. (Euro 15k) Funded by VWS**,** Virtuele Internationale Samenwerking grant. VIS 22029. Main applicant

2021 - 2022
Opportunity map en meetinstrumenten voor robot-kind relaties. (Euro 30k) Funded by SIA, NWO. KIEM.K21.01.103 co-applicant (1 of 3)

2021 - 2023
SOROCOVA: Sociale Robotica ter ondersteuning van COVID-19 LeerAchterstanden. (Euro 150k) Funded by regieorgaan SIA, NWO. Grant number: SIA.COV.01.006. Co-applicant (1 of 3)

2017 - 2022
Implementing robot tutors in a morally justified way. (Euro 180k) Funded by Promotiebeurs voor leraren, NWO. Grant number: 023.010.066/3661. Main applicant

2021
Computable Award 2021 with the robot storytelling cuddle, SAMbuddy

2021
Social robots for reducing mathematics hiatuses in primary education, an exploratory field study (**Outstanding paper Award**) 34th Bled eConference

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[1] Department ranks #10 in "Communication" in the 2020 Shanghai ranking.

2020 Nominated by the Research Centre for Learning and Innovation (LENI) for **HU Researcher of the year** (with the final 8)

Academic Publications and Presentations

Hoi Ki, C.T., Smakman, M.H.J., de Haas, M., van den Berghe, R. (accepted) L2 Vocabulary Learning Through Lexical Inferencing Stories With a Social Robot. *HRI 2023 : International Conference on Human-Robot Interaction*, Stockholm, Sweden

Smakman MHJ, Vanegas DFP, Smit K, Leewis S, Okkerse Y, Obbes J, Uffing T, Soliman M, van der Krogt T, Tönjes L. A (2022) Trustworthy Robot Buddy for Primary School Children. *Multimodal Technologies and Interaction*. 2022; 6(4):29.

Elloumi, L., Bossema, M., De Droog, S. M., Smakman, M. H., Van Ginkel, S., Ligthart, M. E., ... & Allouch, S. B. (2022, August). Exploring requirements and opportunities for social robots in primary mathematics education. In *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)* (pp. 316-322). IEEE.

Smakman, M. H.J., Konijn, E. A., & Vogt, P. A. (2022). Do Robotic Tutors Compromise the Social-Emotional Development of Children?. *Frontiers in Robotics and AI*, *9*.

Broekens, J., Smakman, M. H. J., Erasmus, M., Pols, J., Schimpf M., Sliep, M. Humanoid robots for visually impaired children: towards inclusion of this forgotten group in *ACM/IEEE International Conference on Human-Robot Interaction: Inclusive HRI Workshop: Equity and Diversity in Design, Application, Methods, and Community.* At: Online (Originally Sapporo, Hokkaido, Japan)

Smakman, M., Vogt, P.A. , Konijn E.A. (2021) Moral Considerations on Social Robots in Education: A Multi-stakeholder Perspective. *Computers & Education*, *174*, 104317. https://doi.org/10.1016/j.compedu.2021.104317

Smakman, M.H.J., Smit, K., Buser, L., Monshouwer, T., van Putten, N., Trip, T., Schoof, C., Preciado, D.F., Konijn, E.A., van der Roest, E.M., Tiel Groenestege, W.M. (2021) Mitigating Children's Pain and Anxiety during Blood Draw Using Social Robots. *Electronics*, 10, 1211. https://doi.org/10.3390/electronics10101211

Smakman, M.H.J., Konijn, E.A., Vogt, P., Pankowska, P. (2021) Attitudes towards Social Robots in Education: Enthusiast, Practical, Troubled, Sceptic, and Mindfully Positive. *Robotics*, 10, 24. https://doi.org/10.3390/robotics10010024

Smakman, M.H.J., Smit, K.,Lan, E., Fermin, T., Van Lagen, J. Maas, J., Van Vliet, D., Leewis, S. (2021) *Social Robots for Reducing Mathematics Hiatuses in Primary Education, an Exploratory Field Study* in proceedings of 34th Bled eConference Digital Support from Crisis to Progressive Change. https://doi.org/10.18690/978-961-286-485-9

Smakman, M., Konijn, E.A. (2021) Harms and Benefits of Robot Tutors: A Systematic Literature Review. Under review

Smakman, M., Berket, J., & Konijn, E. A. (2020, October). The Impact of Social Robots in Education: Moral Considerations of Dutch Educational Policymakers. In *2020 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)* (pp. 647-652). IEEE.

Van Ewijk, G., Smakman, M., & Konijn, E. A. (2020, June). Teachers' perspectives on social robots in education: an exploratory case study. In *Proceedings of the Interaction Design and Children Conference* (pp. 273-280).

Konijn, E.A., Smakman, M. & van den Berghe, R. (2020). Use of Robots in Education. In: van den Bulck, J., Sharrer, E., Ewoldsen, D. & Mares, M-L. (Eds). *The International Encyclopedia of Media Psycholog*y. Wiley Publisher

Smakman, M., Jansen, B., Leunen, J., & Konijn, E. (2020) Acceptable Social Robots in Education: A Value Sensitive Parent Perspective. In *INTED2020 Proceedings* (pp 7946-7953).

Smakman M., Konijn E.A. (2020) Robot Tutors: Welcome or Ethically Questionable?. In: Merdan M., Lepuschitz W., Koppensteiner G., Balogh R., Obdržálek D. (eds) *Robotics in Education. RiE 2019.* Advances in Intelligent Systems and Computing, vol 1023. Springer, Cham

Van Liempd D., Sjouw A., Smakman, M., Smit, K. (2019) Social Engineering as an Approach for Probing Organizations to Improve IT Security: A Case Study at a Large International Firm in the Transport Industry. In Kommers, Ravesteijn, Ongena and Isaías (Eds) *IADIS International Conference e-Society* 2019, pp 119-126 .

Goudzwaard, M., Smakman, M., & Konijn, E. A. (2019). Robots are Good for Profit: A Business Perspective on Robots in Education. In *Proceedings of 2019 Joint IEEE 9th International Conference on Development and Learning and Epigenetic Robotics (ICDL-EpiRob)*, 54–60. https://doi.org/10.1109/DEVLRN.2019.8850726

Smakman, M., Konijn, E.A. (2019, February) *Moral Considerations Regarding Robots in Education: A Systematic Literature Review*. Paper presented at Etmaal van de Communicatiewetenschap, 7-8 February 2019. Nijmegen, The Netherlands

Smakman, M., Konijn. E.A. (2018, December) *Considerations on moral values regarding robot tutors.* Presented at the Symposium on Robots for Language Learning. 12-13 December 2018. Koç University, Istanbul, Turkey

Smakman, M. (2018, February). *Moral concerns regarding robot tutors, a review.* Poster presented at the ATEE 2018 Winter Conference - Technology and Innovative Learning, Utrecht, The Netherlands.

Dignum V., Caon, M. Chatila, R., Dennis, L. Génova, G., López-Sánchez, M. Pavón, J., Slavkovik, M., Smakman, M., Van Steenbergen, M. Van Der Torre, L., Villata, S., De Wildt, T. (2018, December). Ethics by Design: necessity or curse?. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 60-66). ACM.

Smakman, M. (2017, October). *Robot Tutor and Moral Concepts*. Paper presented at Thematic day Ethics by Design at PRIMA2017, Nice, France.

Smakman, M. (2016, October). *Robots and Moral Obligations.* Paper presented at International Research Conference Robophilosophy 2016 / TRANSOR 2016, Aarhus, Denmark.

Smakman, M. (2016, July). *Robots and Moral Obligations.* Paper presented at Designing Moral Technologies -Theoretical, Practical and Ethical Issues, Ascona, Switzerland.

Smakman, M. (2016). Robots and moral obligations. In Seibt, J., Nørskov, M., Schack Andersen, S. (Ed.) *What Social Robots Can and Should Do: Proceedings of Robophilosophy 2016 / TRANSOR 2016*, 184-189. dio 10.3233/978-1-61499-708-5-184.

Professional Presentations, Publications en Nieuwsberichten

Smakman, M.H.J. (12-01-2023) Vermelding op Jeugdjournaal robots in het Wilhelmina Kinderziekenhuis. NOS Jeugdjournaal. URL: https://www.npostart.nl/nos-jeugdjournaal/12-01-2023/POW_05468870

Smakman, (11-01-2023) Story gedeeld door NOS over robots tegen prikangst, NOS, URL: https://nos.nl/l/2459480

Smakman, M.H.J. (10/01/23**)** Radio Interview Robots tegen prikangst, Radio M Utrecht, https://www.rtvutrecht.nl/radio/aflevering/utrecht-is-wakker/3511531_20230110080000

Smakman, M.H.J. (10-01-2023) TV Interview Robots tegen prikangst, U nieuws, RTV Utrecht. https://www.rtvutrecht.nl/tv/aflevering/unieuws/RTVU_3510114_20230110170000

Smakman, M. H. J., (2022). Een robot helpt kinderen hun hart te luchten. Hogeschool Utrecht. https://www.hu.nl/hustories/sambuddy

Smakman, M., Leewis S. (2022). *Human-Robot Interaction*. EHV Innovation Café.

Smakman, M.H.J. (2022) *Inspiratie Talk: waarom mensen zich hechten aan humanoïde robots*. Vegro Zorgevent in LAB39, Katwijk

Smakman (2021, June) *Genomineerde kinderknuffelrobot sambuddy straks ook maatje voor ouderen.* Interview by Gerard Rutten, Studium Generale. https://trajectum.hu.nl/genomineerde-kinderknuffelrobot-sambuddy-straks-ook-maatje-voor-ouderen/

Smakman (2021, May) Sociale robots inzetten in het primair onderwijs? Deze waarden verdienen aandacht. Interview by Kennisnet. https://www.kennisnet.nl/artikel/10160/sociale-robots-inzetten-in-het-primair-onderwijs-deze-waarden-verdienen-aandacht/

Konijn, E.A. and Smakman, M. (20-01-2020) *Zorgrobots* presented at Hersenziekten, Kunnen data je leven redden? UMC Utrecht & NewScientist. Tivoli Vredenburg, Utrecht, The Netherlands

Smakman, M. (2019) De robotdocent komt eraan, maar hoe? *AG Connect.* Januari/ Februari 2019. pp 70-73

Smakman, M. and Konijn, E.A. (2019-02-07) *Onderwijsrobots: van harte welkom of ethisch onverantwoord?* Presented at Robots en AI in het onderwijs. Den Haag, The Netherlands.

Smakman, M. And Konijn, E.A. (2019-01-31) *Moral challenges and opportunities for educational robots* Presented at Workshop How do we work with educational robots? De Waag, Amsterdam,The Netherlands.

Smakman, M. (2018, May*) Strategic workshop on Privacy and Security*. Presented at PON heidagen,Westhem, The Netherlands

Smakman, M. (2018, May) *Robotics & Ethics: Ethical influences of Technology on clients and society*. Presented at Deloitte Consulting, Amsterdam, The Netherlands

Bouwers, A., de Bruin, J., Duong, R., Smakman, M. (2018, February) *Shoppen volgens China.* Symposium conducted at the VPRO-Tegenlicht Meet Up. Utrecht, The Netherlands

Smakman, M. (2017, October). *Lectures and Workshops on Working with e-learning.* Presented at Ho Chi Minh City University of Natural Resources and Environment, Ho Chi Minh, Vietnam.

Smakman, M. (2017, October). *Privacy and Innovation: Practical and Philosophical Issues.* Presented at, CMS Compliance Academy, Utrecht, The Netherlands

Smakman, M. (2017, March*). The General Data Protection Regulation, the right way?.* Presented at Talking Dinner Inergy, Utrecht, The Netherlands

Smakman, M. (2017, March). *Big Data and Ethics.* Presented at Young Employee Schneider Event, Schneider Electric, Hoofddorp, The Netherlands

Smakman, M. (2016, November). *Lectures and Workshops on Smart Sustainable Cities and e-learning.* Presented at Danang University of Science and Technology, Da Nang, Vietnam.

Smakman, M. (2014, April). *Business success and Corporate Social Responsibility*. Presented at Make A Difference Event, Erasmus University Rotterdam, Rotterdam, The Netherlands.